

EFT Theory Uncertainties at the LHC

Presented by F. Montagno

In collaboration with S. Chang, M. Luty, T. Ma and A. Wulzer.

Outline

Motivation

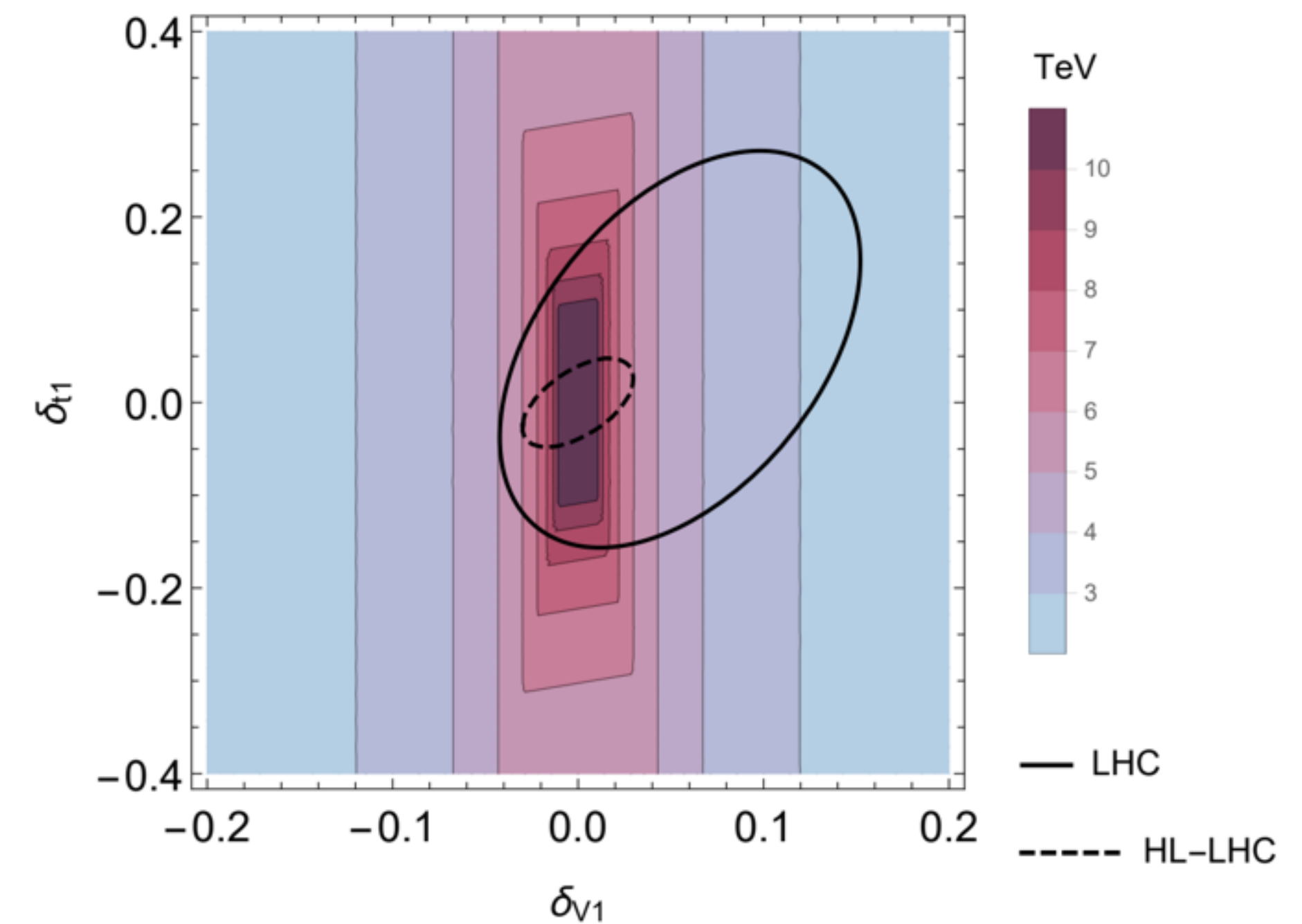
1. EFTs and their validity
2. The Problem: EFT Validity
3. The Clipping Proposal
4. Our Proposal

Approach

5. Low Energy Expansion
6. Parametrizing EFT Uncertainties
7. Toy Model Implementation
8. Comparison with UV Models

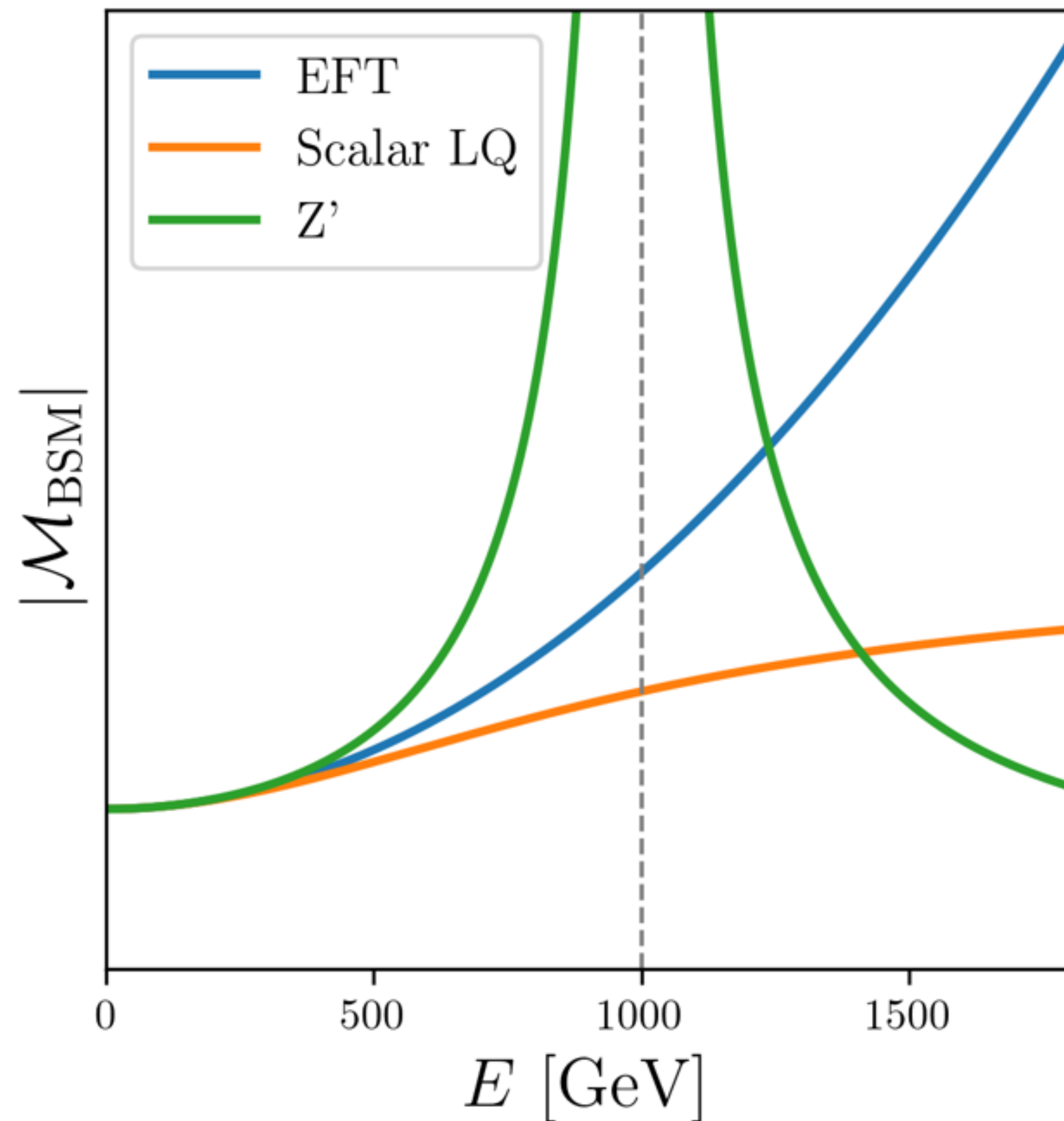
EFTs and their validity

- EFT is a low-energy parametrization of UV models.
Useful for model-independent searches.
- Precision measurements $\rightarrow$
Constraints on Wilson coefficients.
- Under which conditions is the EFT reliable?



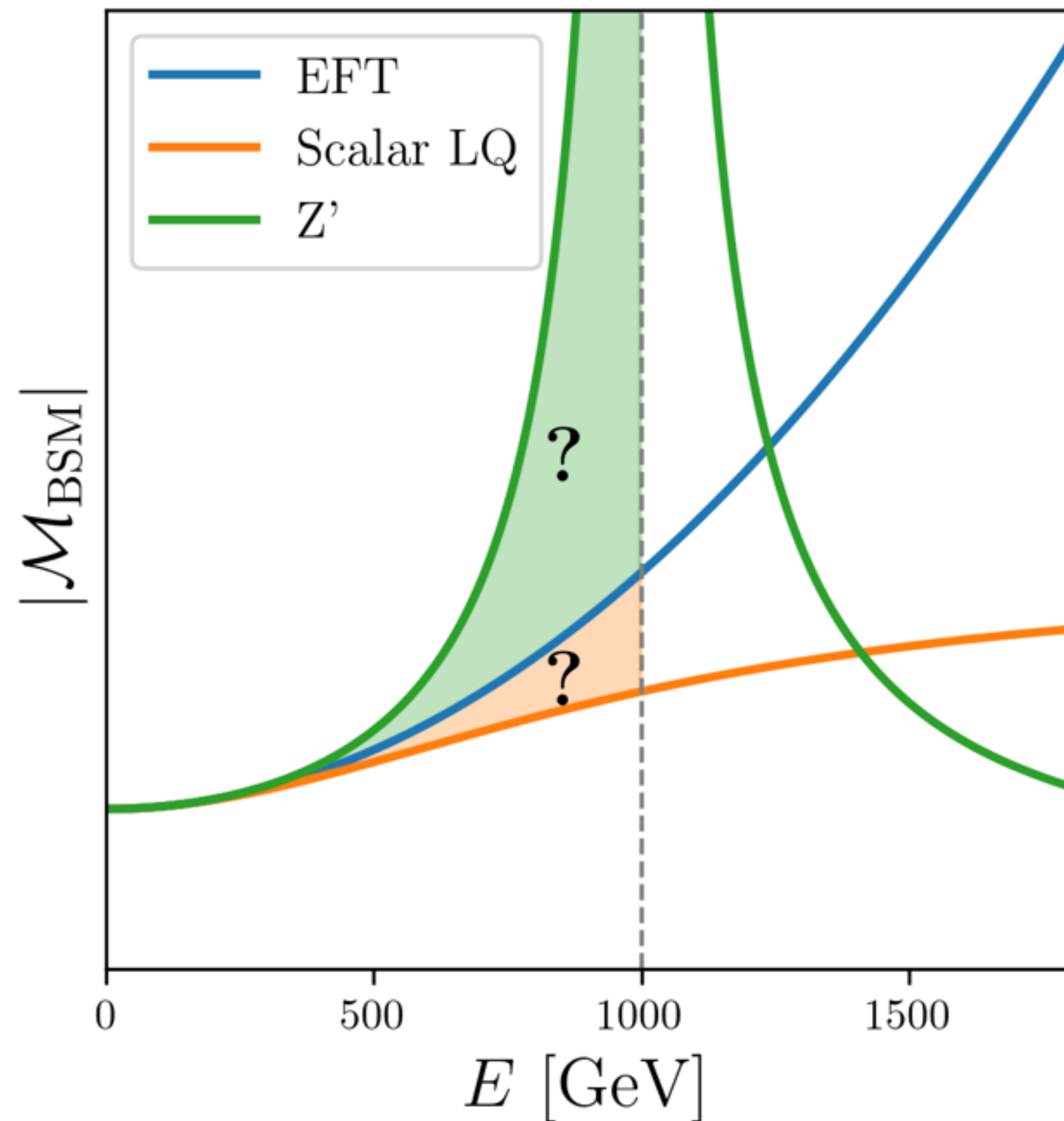
Abu-Ajamieh, Chang, Chen, Luty (2020)

The Problem: EFT Validity



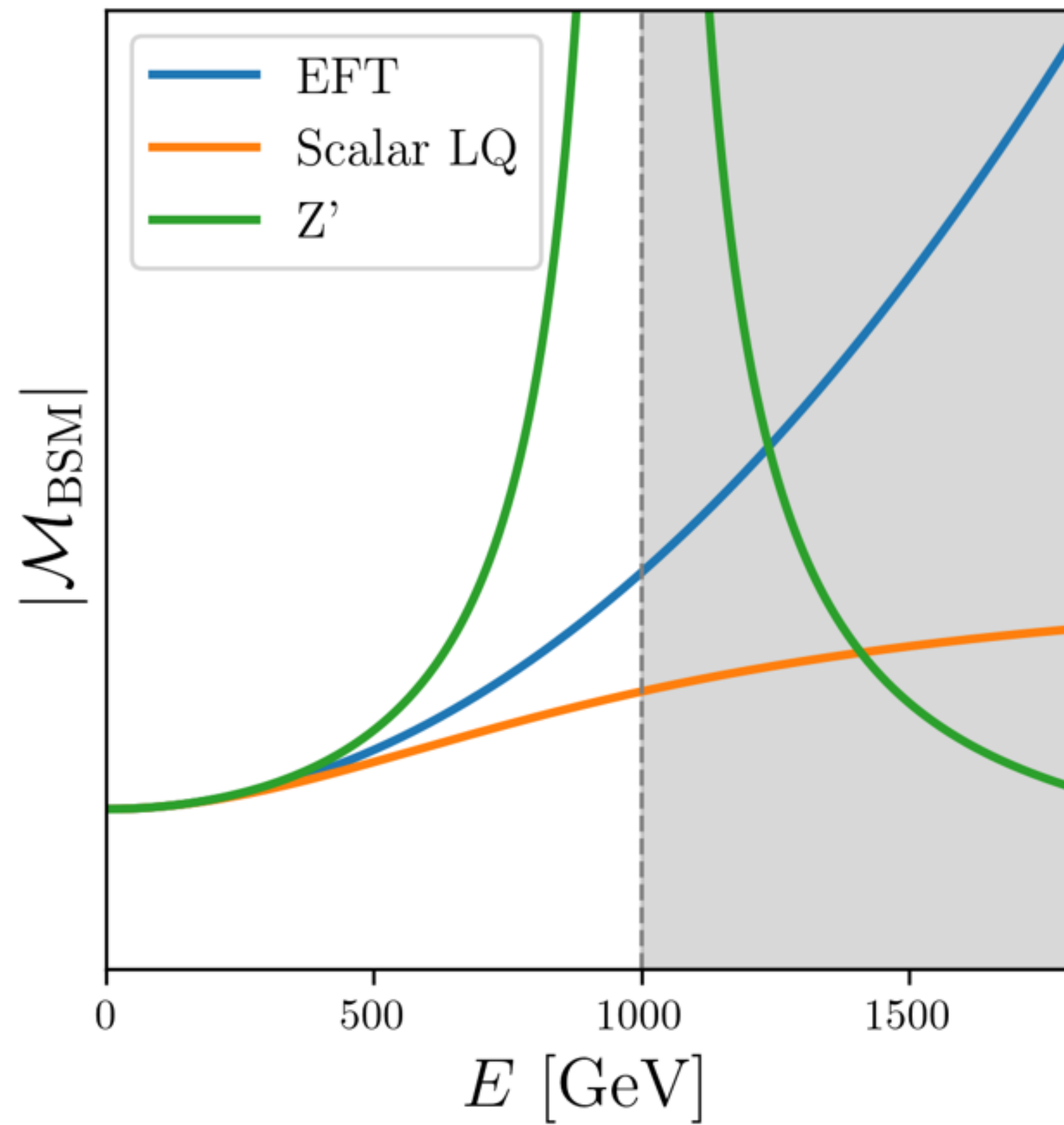
- An EFT describes the effect of heavy particles with mass M in an expansion in E/M
- At $E \gtrsim M$ the EFT breaks down
- Data in the high-energy regime ($E \gtrsim M$) could give **too strong** constraints

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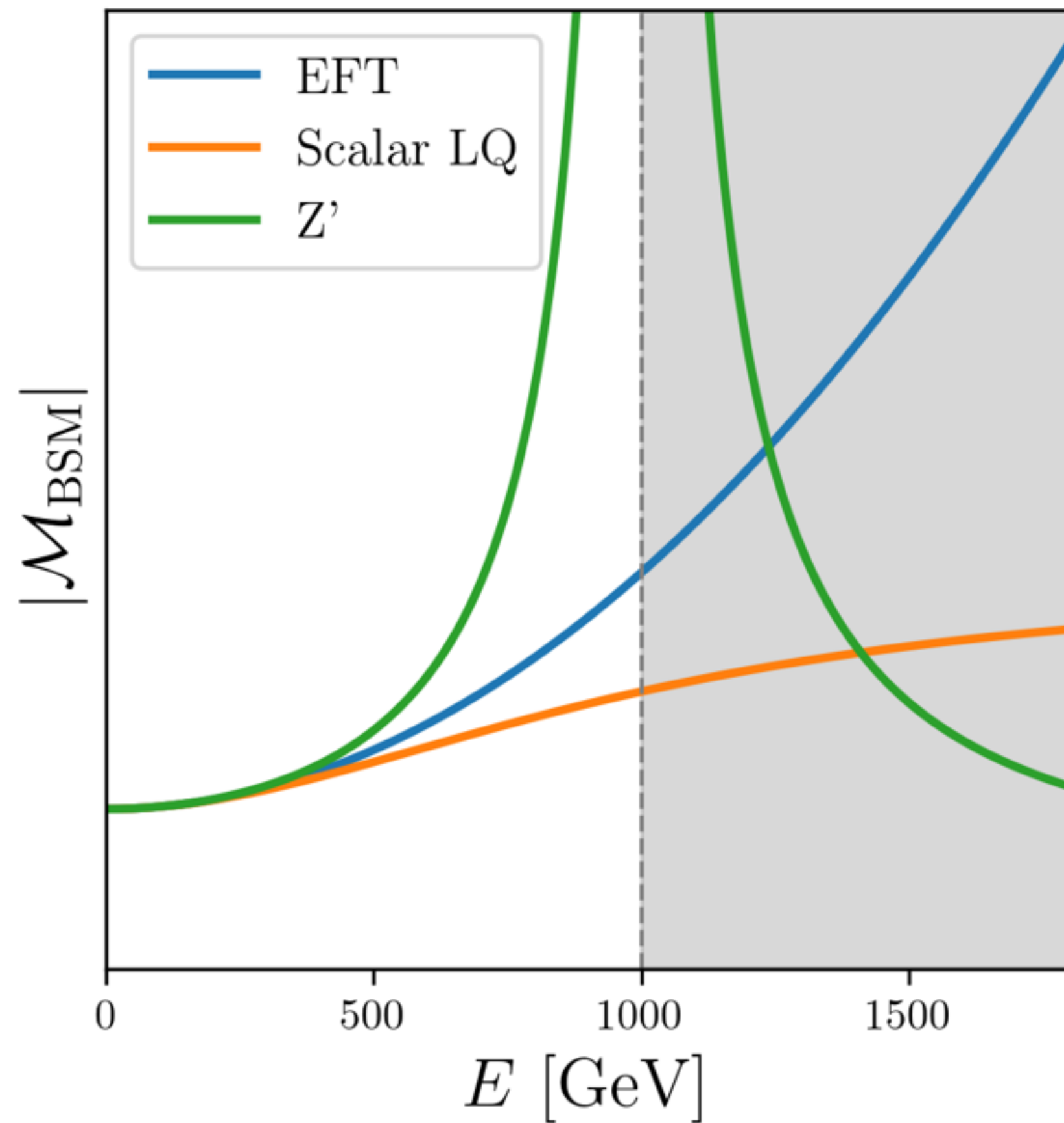
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The Clipping Proposal



- Don't use data above a cutoff $E_{\text{cut}} < M$

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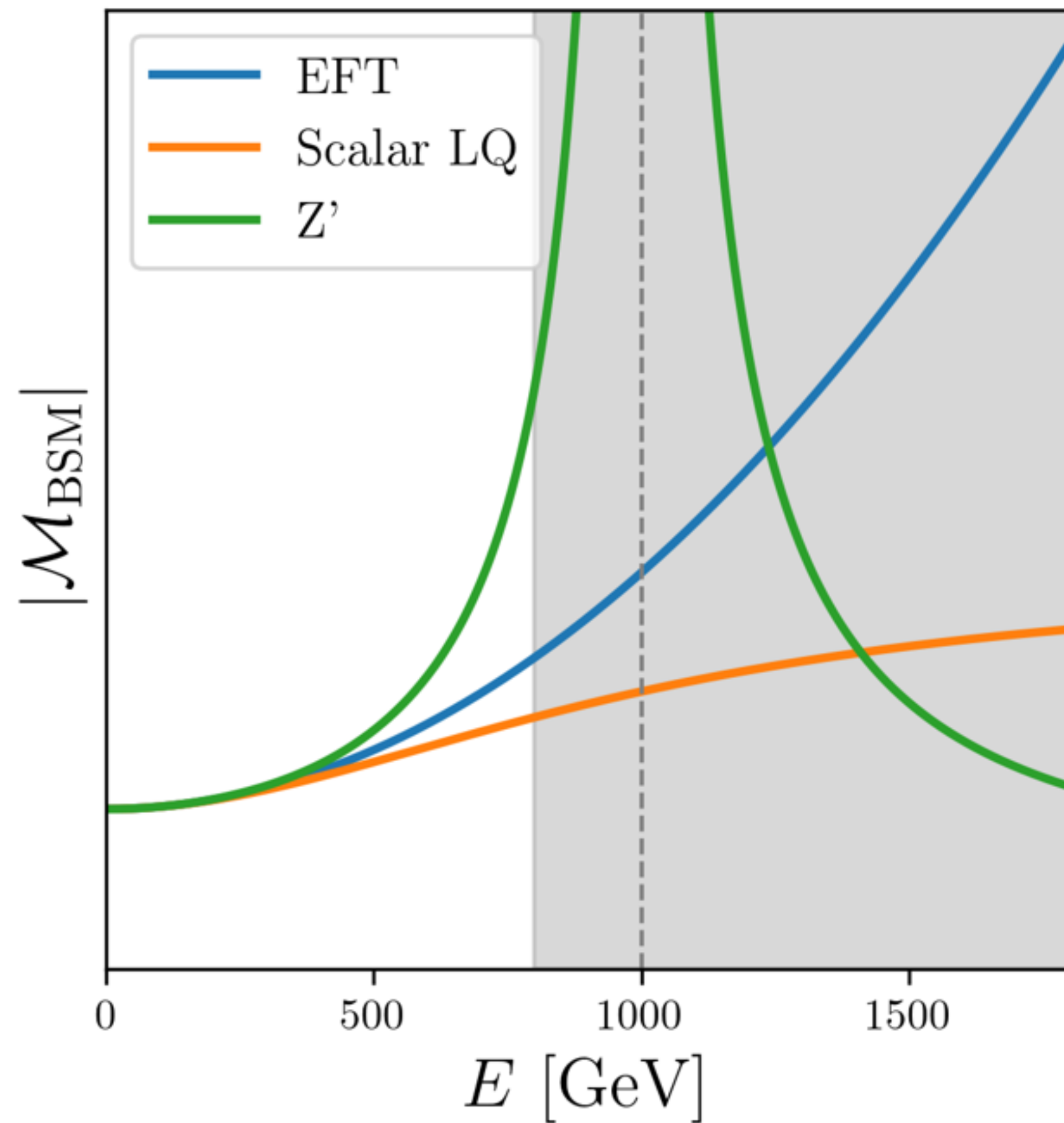


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Problems

- Where to cut?
- Can I reconstruct ' E '?

The Clipping Proposal



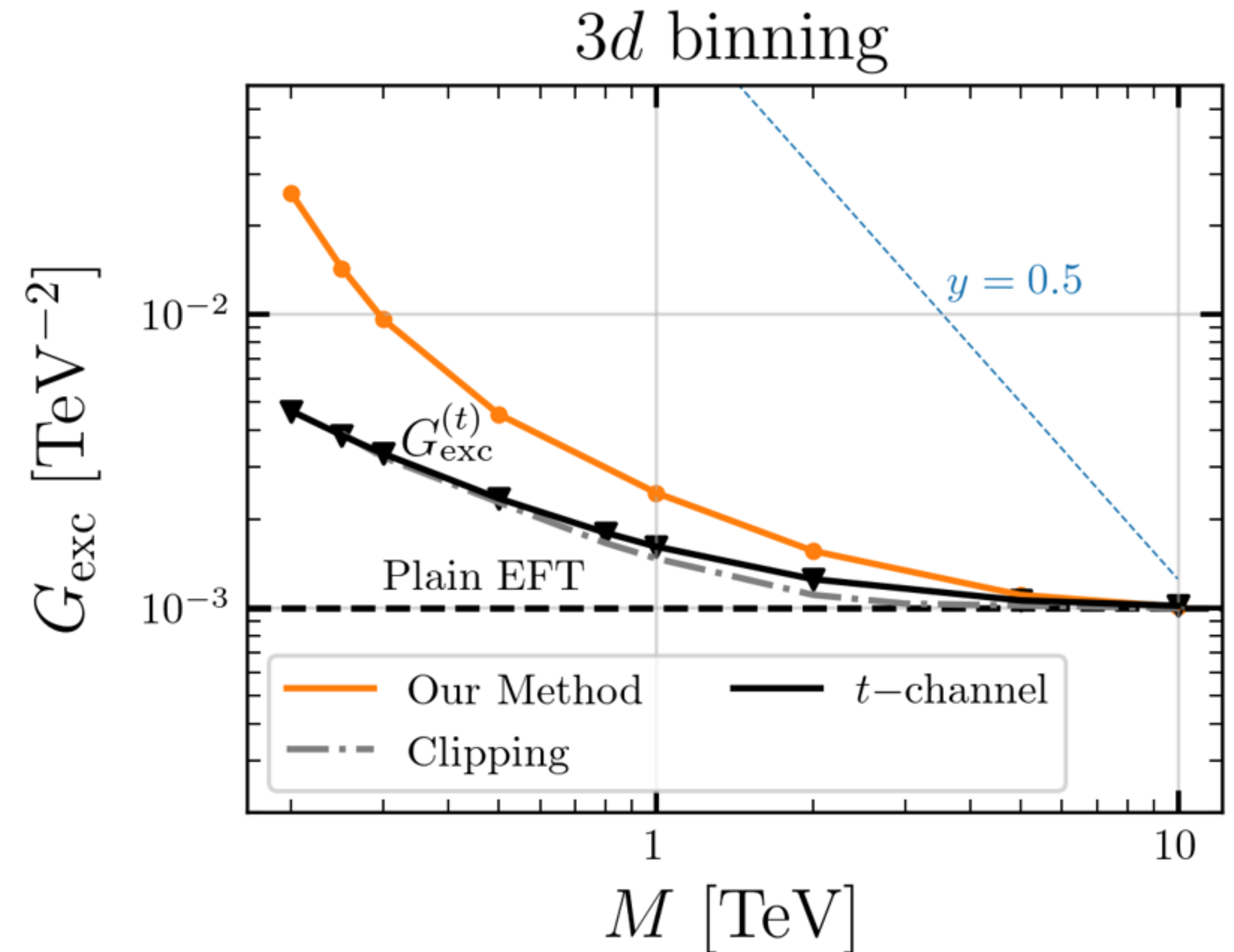
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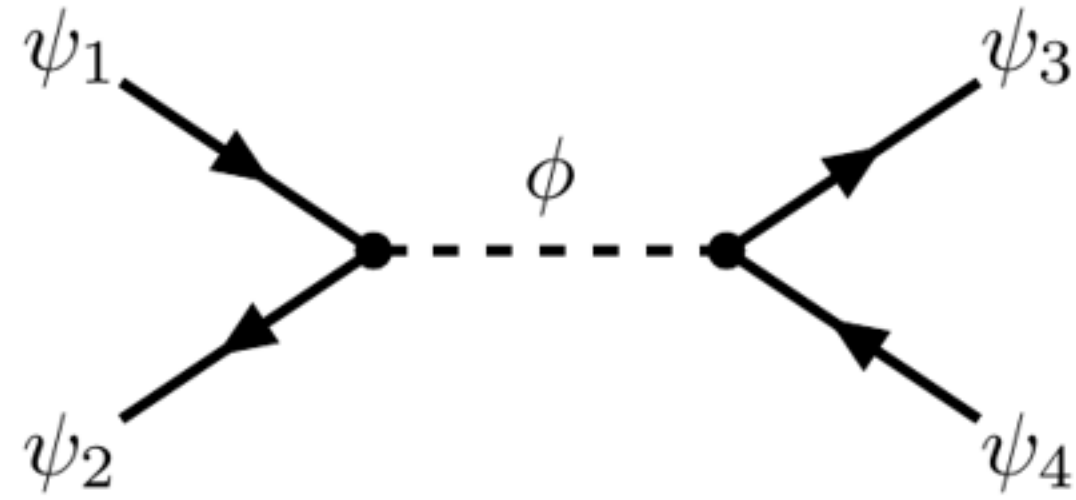
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Our Proposal

- Introduce parameterized uncertainties into EFT predictions via **nuisance parameters** that reflect the higher order corrections.
- The scale M at which the EFT **breaks down** is treated as an independent parameter.
- EFT uncertainty $\longleftrightarrow$ ignorance about UV
Unavoidable prior assumptions
about the size of allowed deviations.

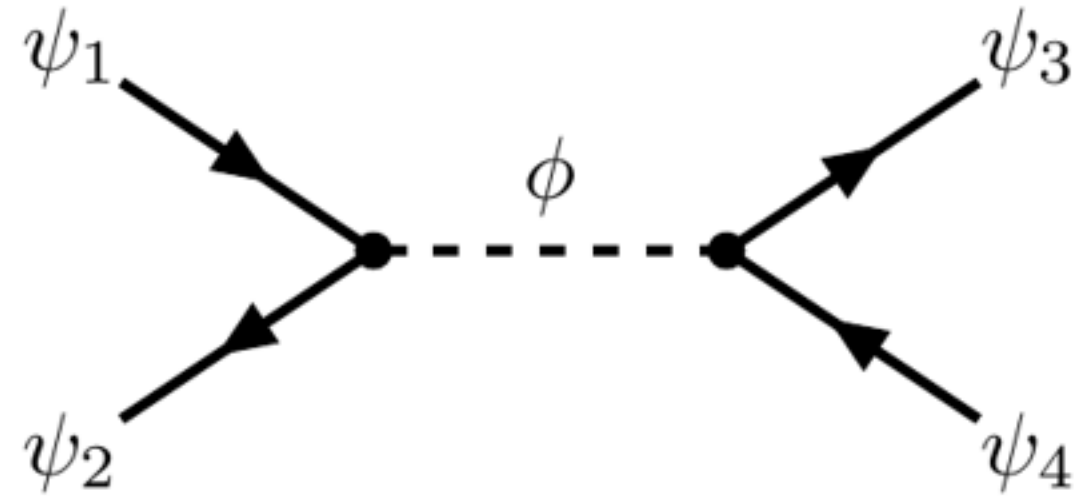


Low Energy Expansion



$$\mathcal{M}_{\text{BSM}} = g^2 \frac{1}{\hat{s} - M^2}$$

Low Energy Expansion

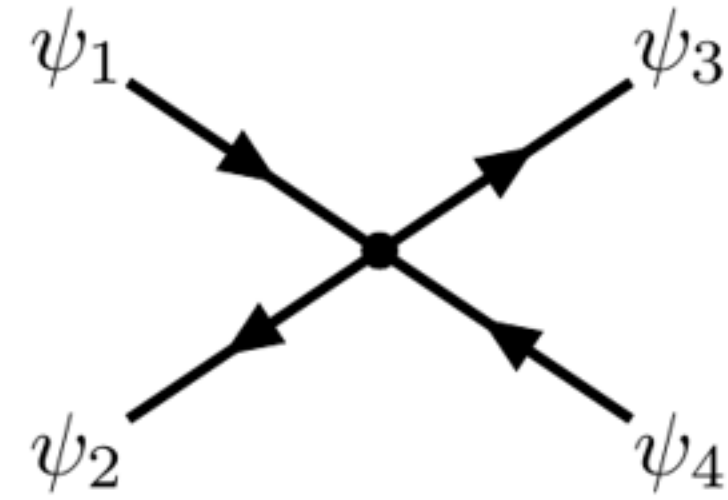


$$\mathcal{M}_{\text{BSM}} = g^2 \frac{1}{\hat{s} - M^2}$$

Integrating out new physics



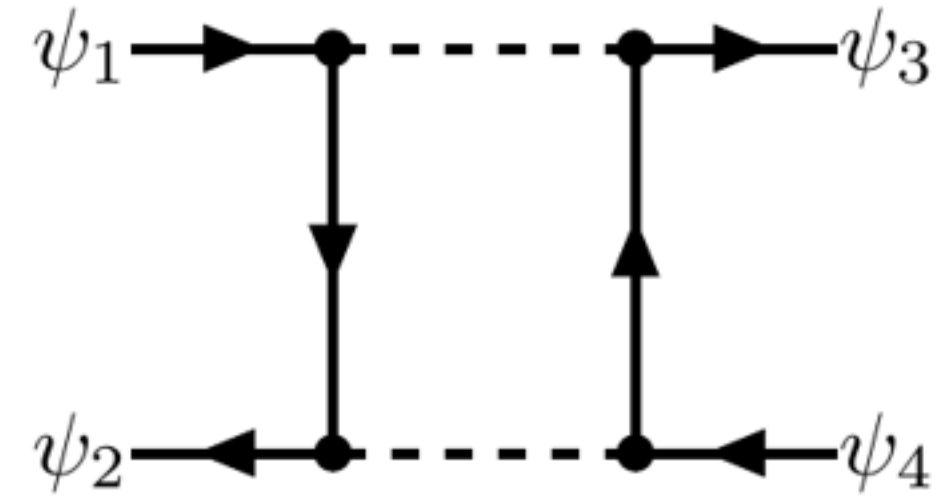
$$\hat{s} \ll M^2$$



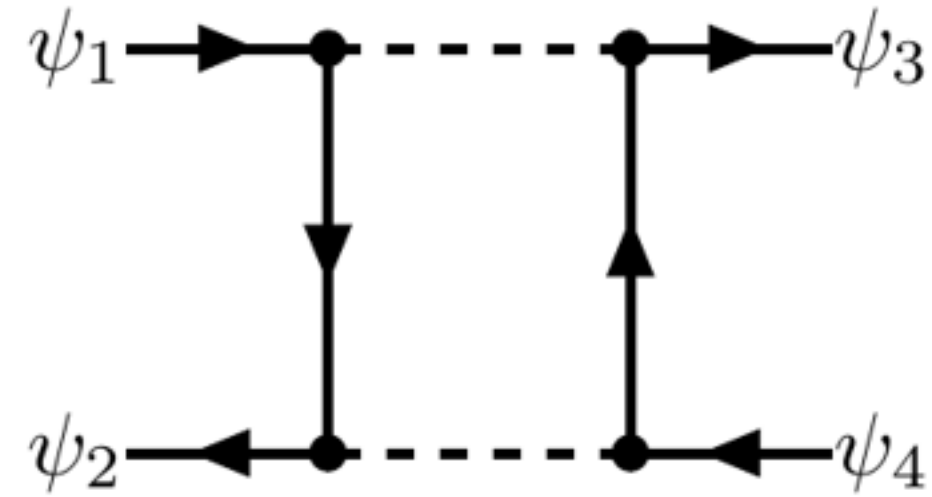
$$\mathcal{M}_{\text{EFT}} = G \left[1 + \underbrace{\frac{\hat{s}}{M^2} + \frac{\hat{s}^2}{M^4} + \dots}_{\text{Mandelstam Descendants}} \right]$$

Mandelstam Descendants

Low Energy Expansion



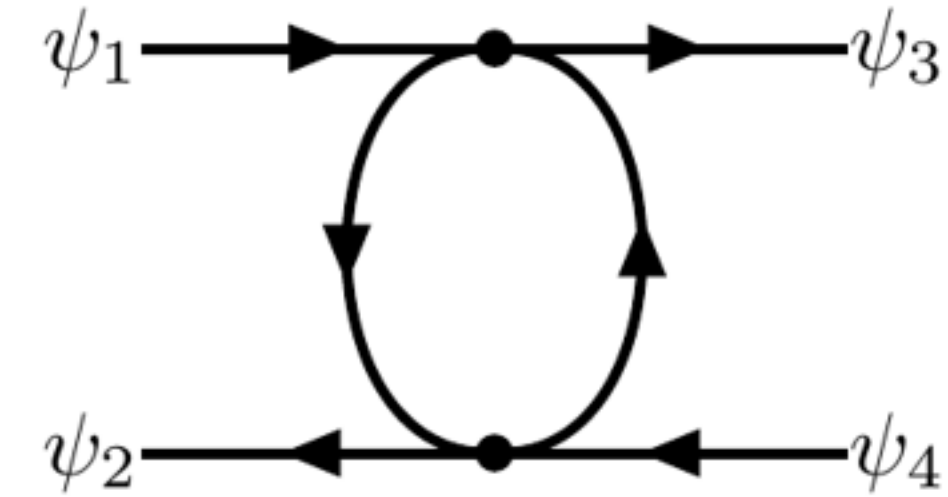
Low Energy Expansion



Integrating out new physics



$$\hat{s}, \hat{t} \ll M^2$$



$$\mathcal{M}_{\text{EFT}} = G \left[1 + \underbrace{c_1 \frac{\hat{s}}{M^2} + c_2 \frac{\hat{t}}{M^2} + c_3 \frac{\hat{s}^2}{M^4} + c_4 \frac{\hat{s}\hat{t}}{M^4} + \dots}_{\text{Mandelstam Descendants}} \right]$$

Mandelstam Descendants

Parameterizing EFT Uncertainties

The general structure from integrating out heavy physics is

$$\mathcal{M}_{\text{BSM}}(12 \rightarrow 34) = \sum_{\mathcal{O}} G_{\mathcal{O}} \mathcal{M}_{\mathcal{O}} \left[1 + c_{\mathcal{O},1} \frac{\hat{s}}{M^2} + c_{\mathcal{O},2} \frac{\hat{t}}{M^2} + c_{\mathcal{O},3} \frac{\hat{s}^2}{M^4} + c_{\mathcal{O},4} \frac{\hat{s}\hat{t}}{M^4} + \dots \right]$$

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What else do we need?

- A prior assumption that enforces $\mathcal{O}(1) \ c_{\mathcal{O},i}$
- A modeling for $E \gtrsim M$ data (*Avoid Clipping*)

Parameterizing EFT Uncertainties

Given a single EFT operator $\mathcal{O}$, with Wilson coefficient G , we probe

$$\mathcal{M} = \mathcal{M}_{\text{SM}} + G\mathcal{M}_{\mathcal{O}} + O(G^2)$$

The **EFT uncertainties** are parameterized by rescaling the Wilson coefficient

$$G \rightarrow G\mathcal{R}_\nu(x_1, x_2, x_3), \quad x_1 = \frac{\hat{s}_{12}}{M^2}, \quad x_2 = \frac{\hat{s}_{13}}{M^2}, \quad x_3 = \frac{\hat{s}_{14}}{M^2}$$

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Important: No event generation needed!

Parameterizing EFT Uncertainties

The rescaling factor should

- **Reproduce the expansion** for $E \ll M$ (No unphysical energy growth)
- Account for **lack of predictivity** for $E \gtrsim M$ (100% theory uncertainty)

$$\mathcal{R}_\nu(x_1, x_2, x_3) = \left(\frac{F(x_{\max})}{x_{\max}} \right)^{p/2} \sum_{\alpha, \beta, \gamma \geq 0} \nu_{\alpha\beta\gamma} F(x_1)^\alpha F(x_2)^\beta F(x_3)^\gamma$$

$$F(x) = \begin{cases} x & |x| \leq 1 \\ \text{sign}(x) & |x| > 1 \end{cases}, \quad \nu_{000} = 1, \quad x_{\max} = \max(x_1, x_2, x_3)$$

Parameterizing EFT Uncertainties

- When $|x_{1,2,3}| \ll 1 \longrightarrow G\mathcal{R}_\nu(x_1, x_2, x_3) \mathcal{M}_\odot = G \mathcal{M}_\odot [1 + \nu_{100} x_1 + \nu_{010} x_2 + \nu_{110} x_1 x_2 + \dots]$

Parameterizing EFT Uncertainties

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This reproduces

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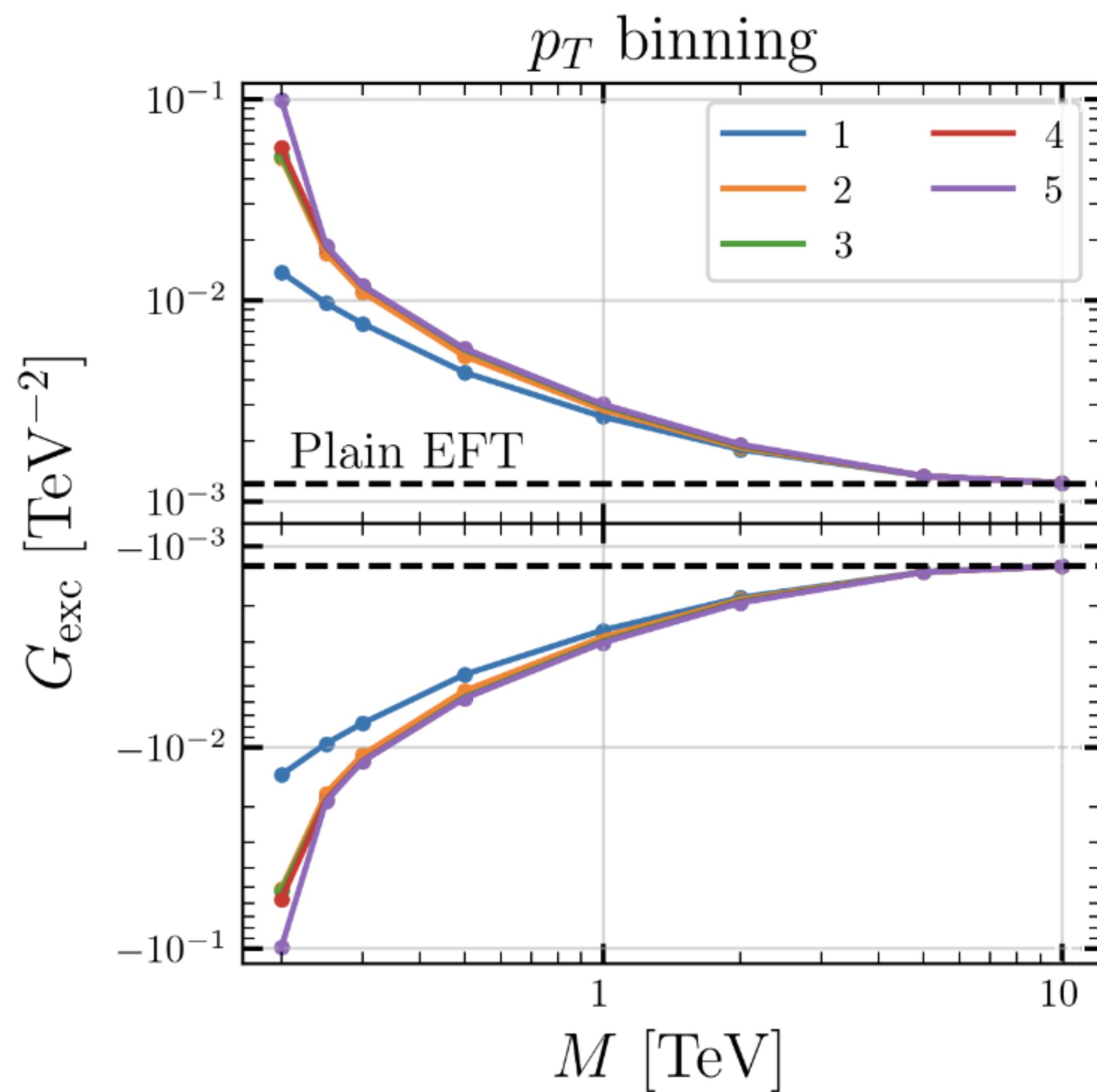
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- In the **high energy limit**, $\mathcal{R}_\nu(x_1, x_2, x_3) \sim E^{-p}[1 + \nu_{100} + \nu_{010} + \nu_{110} + \dots]$

We choose p so that the modified amplitude is **independent of energy** in the UV

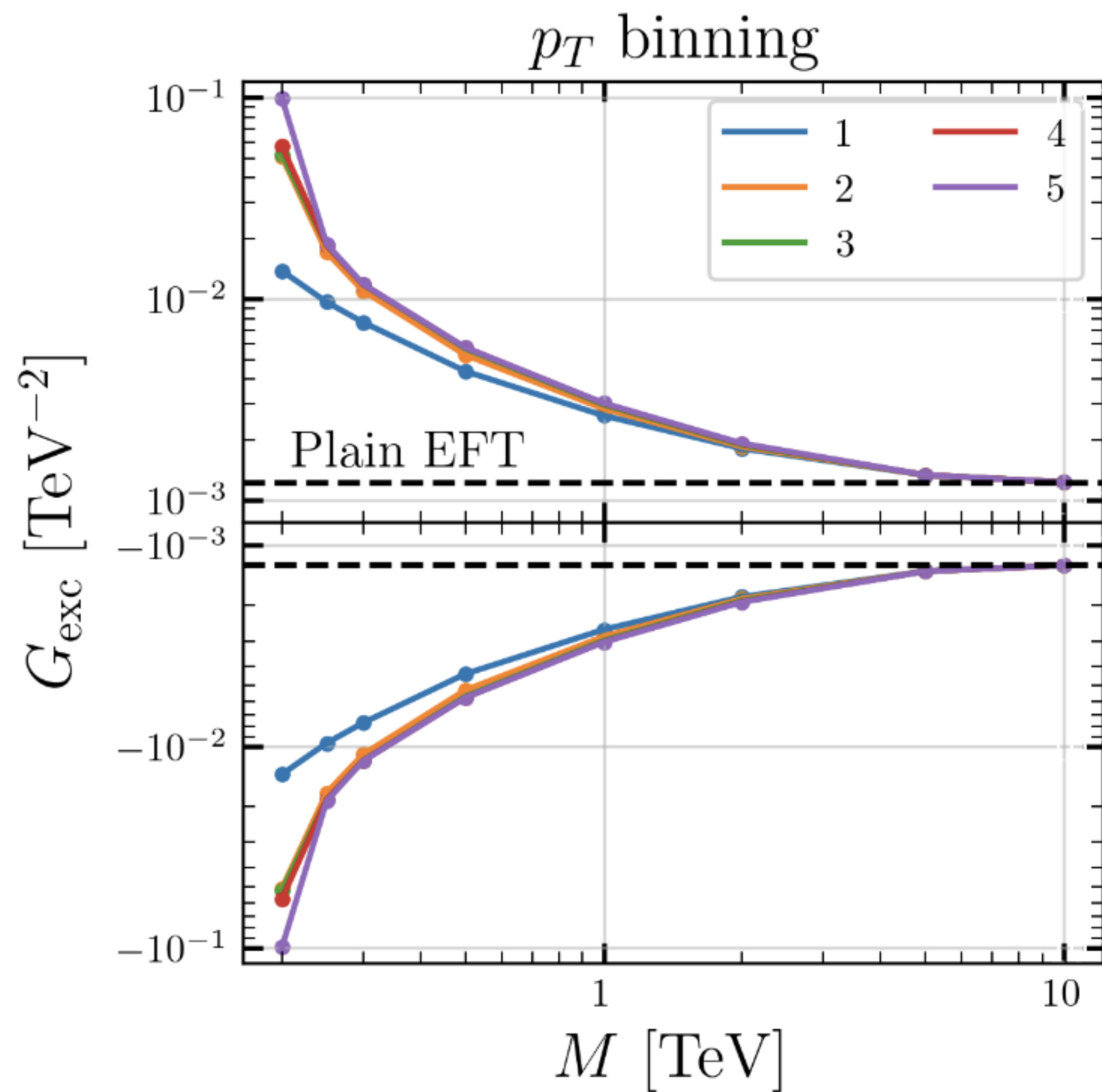
Toy Model Implementation



$$\mathcal{L} = \mathcal{L}_{\text{QED}} + \mathcal{L}_{\text{QCD}} - G(\bar{u}_L \gamma_\mu u_L)(\bar{e}_R \gamma^\mu e_R)$$

Partonic process $u\bar{u} \rightarrow e^+e^-$, energy growth $\sim E^2$

Toy Model Implementation



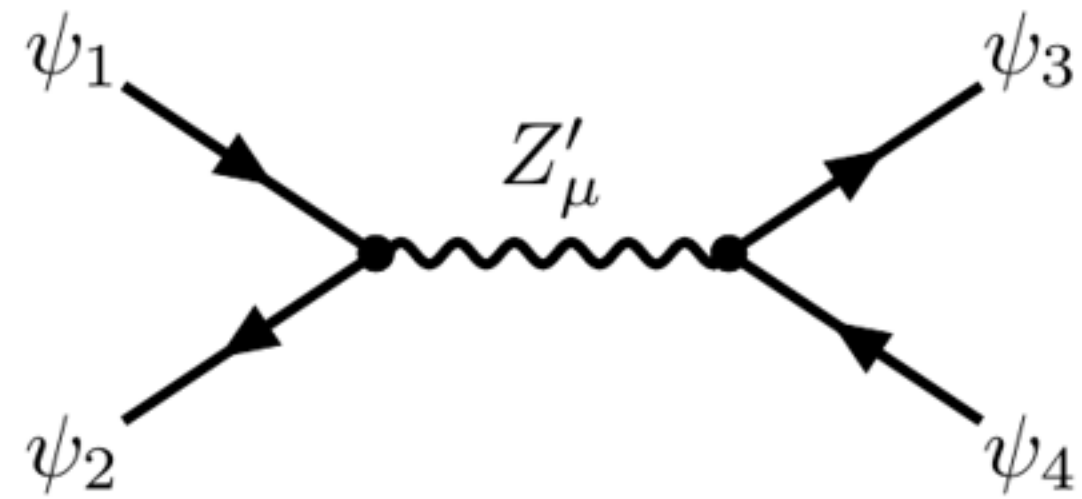
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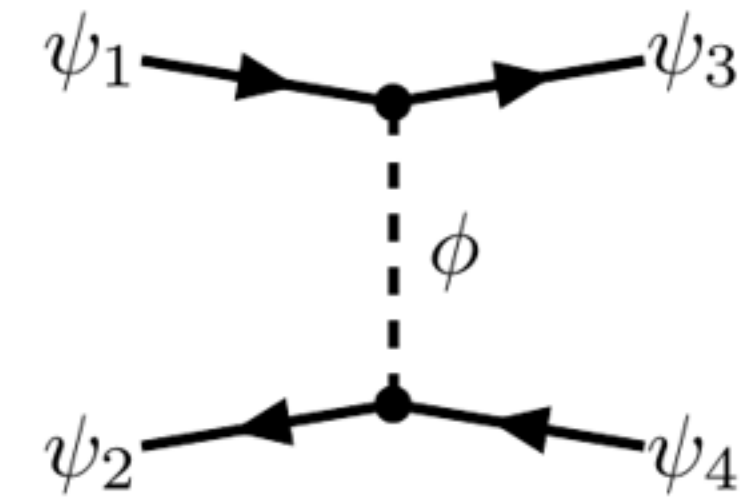
Properties

- $n_\nu \sim 2$ gives convergent results
- Asymptotic behavior as Plain EFT

Comparison with UV Models

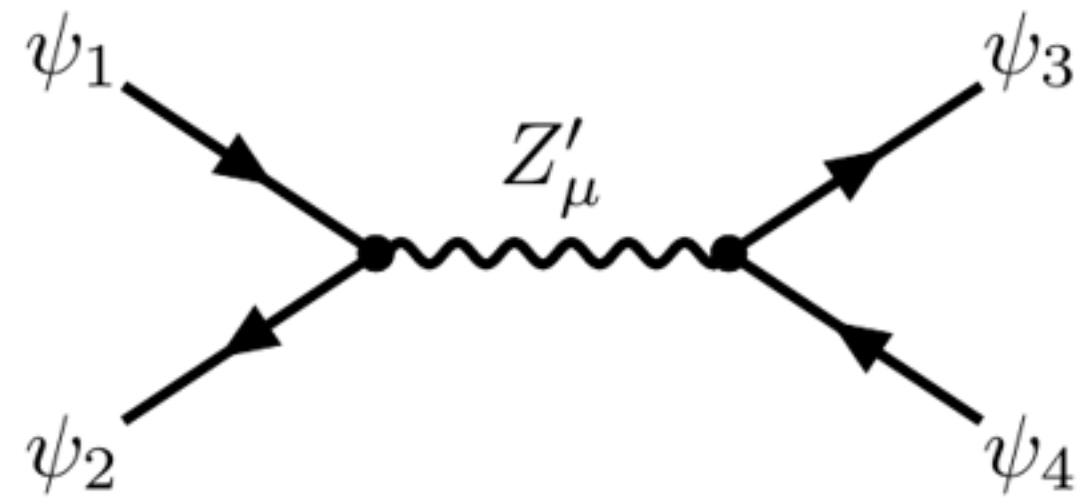


$$\mathcal{L}_{Z'} = gZ'_\mu(\bar{u}_L\gamma^\mu u_L + \bar{e}_R\gamma^\mu e_R)$$

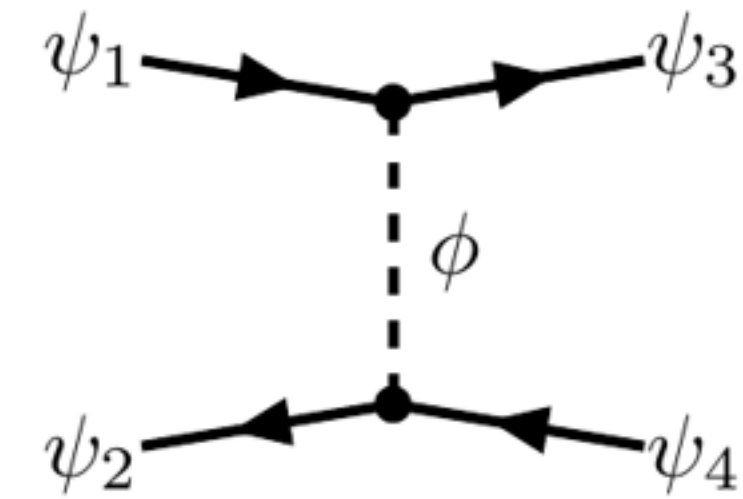


$$\mathcal{L}_\phi = y(\phi\bar{u}_L e_R + \text{h.c.})$$

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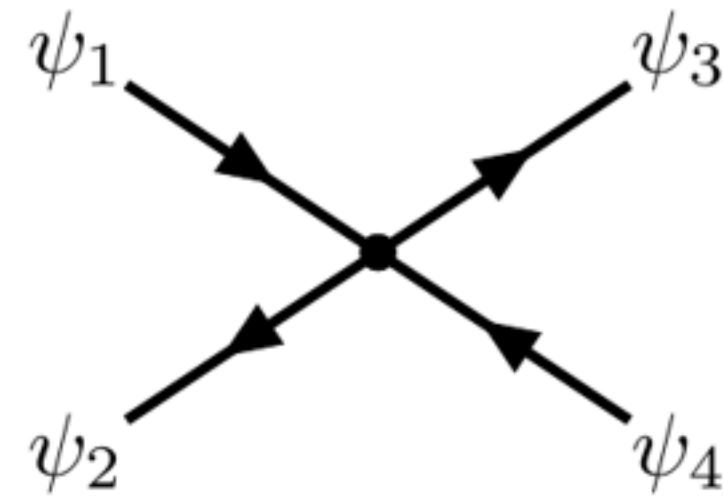


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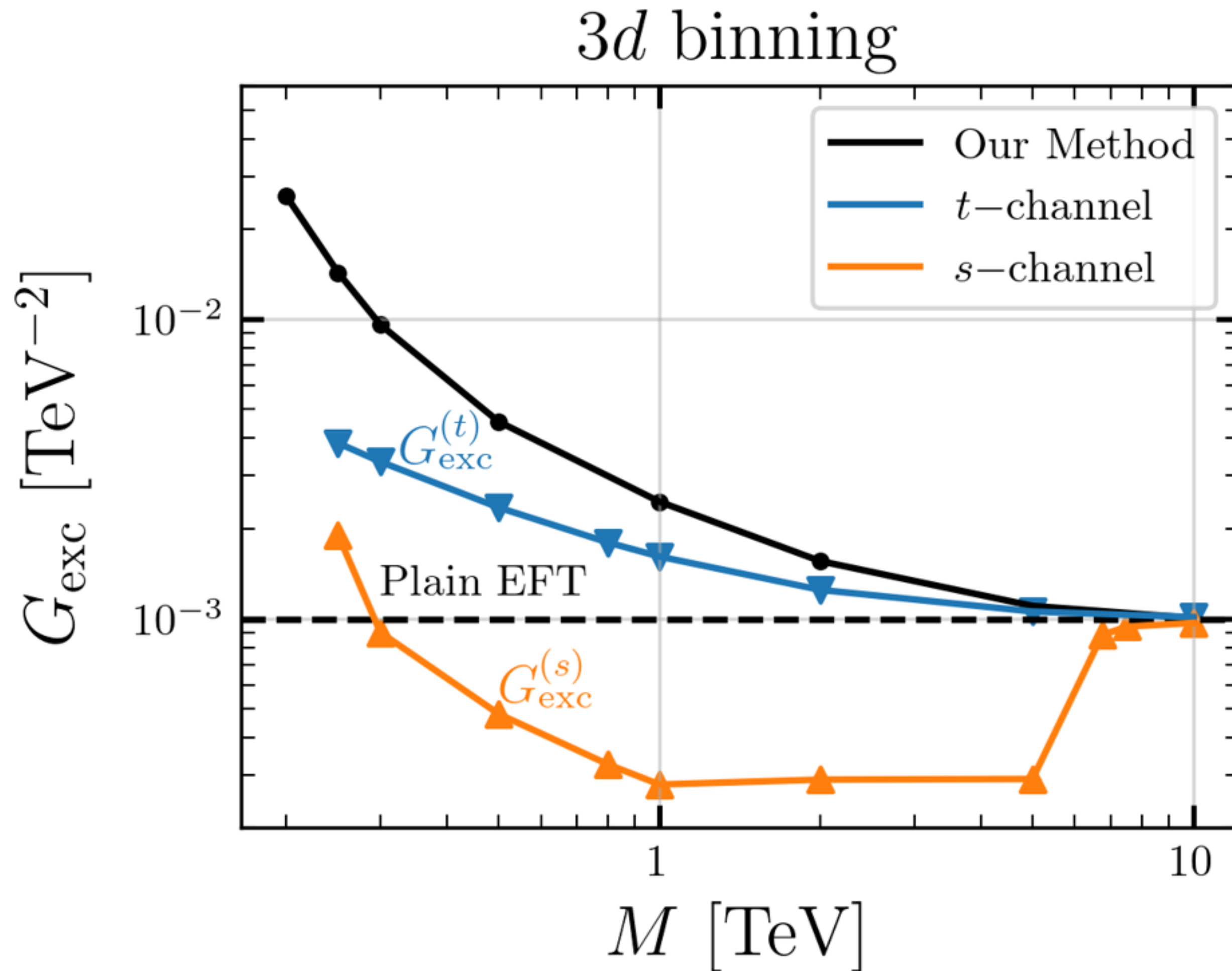
$$G = \frac{g^2}{M_Z^2}$$



$$\mathcal{L}_{\text{EFT}} = -G(\bar{u}_L\gamma_\mu u_L)(\bar{e}_R\gamma^\mu e_R)$$

$$G = \frac{y^2}{2M_\phi^2}$$

Exclusion Limits



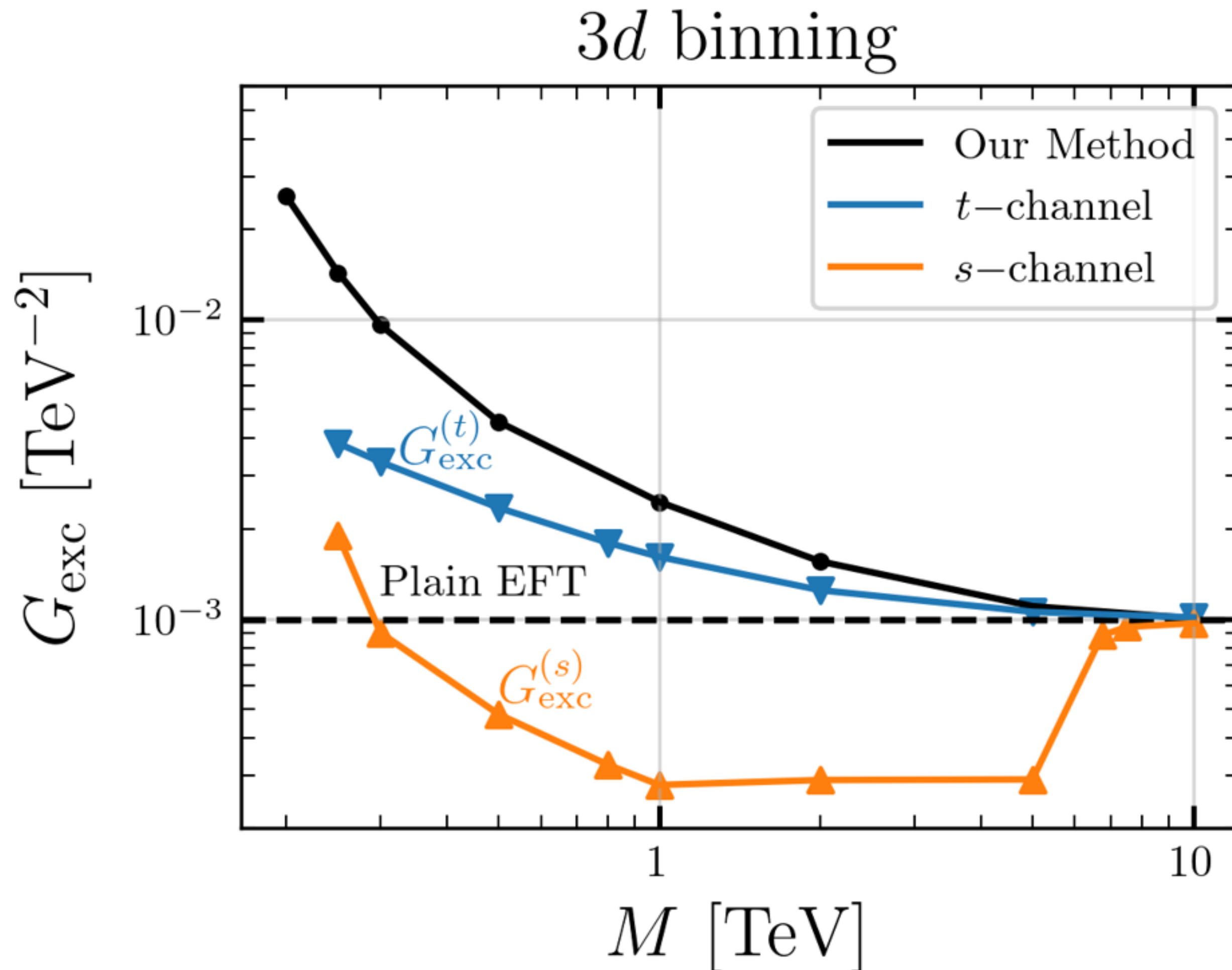
$$\mathcal{L}_{\text{Po}} = \prod_{b \in \text{bins}} \text{Pois}[\text{O}_b | \text{E}_b]$$

$$\mathcal{L}_{\text{prior}}[\nu] = \prod_{\alpha, \beta, \gamma} \exp\left(-\frac{\nu_{\alpha\beta\gamma}^2}{2\sigma_\nu^2}\right), \quad \text{baseline: } \sigma_\nu = 3$$

$$\mathcal{L}[G, \nu; \text{O}] = \mathcal{L}_{\text{Po}}[\text{E}_b(G, \nu); \text{O}] \times \mathcal{L}_{\text{prior}}[\nu]$$

$$t(G) = 2 \log \frac{\mathcal{L}[\hat{G}, \hat{\nu}; \text{O}]}{\mathcal{L}[G, \hat{\nu}_G; \text{O}]}$$

Exclusion Limits



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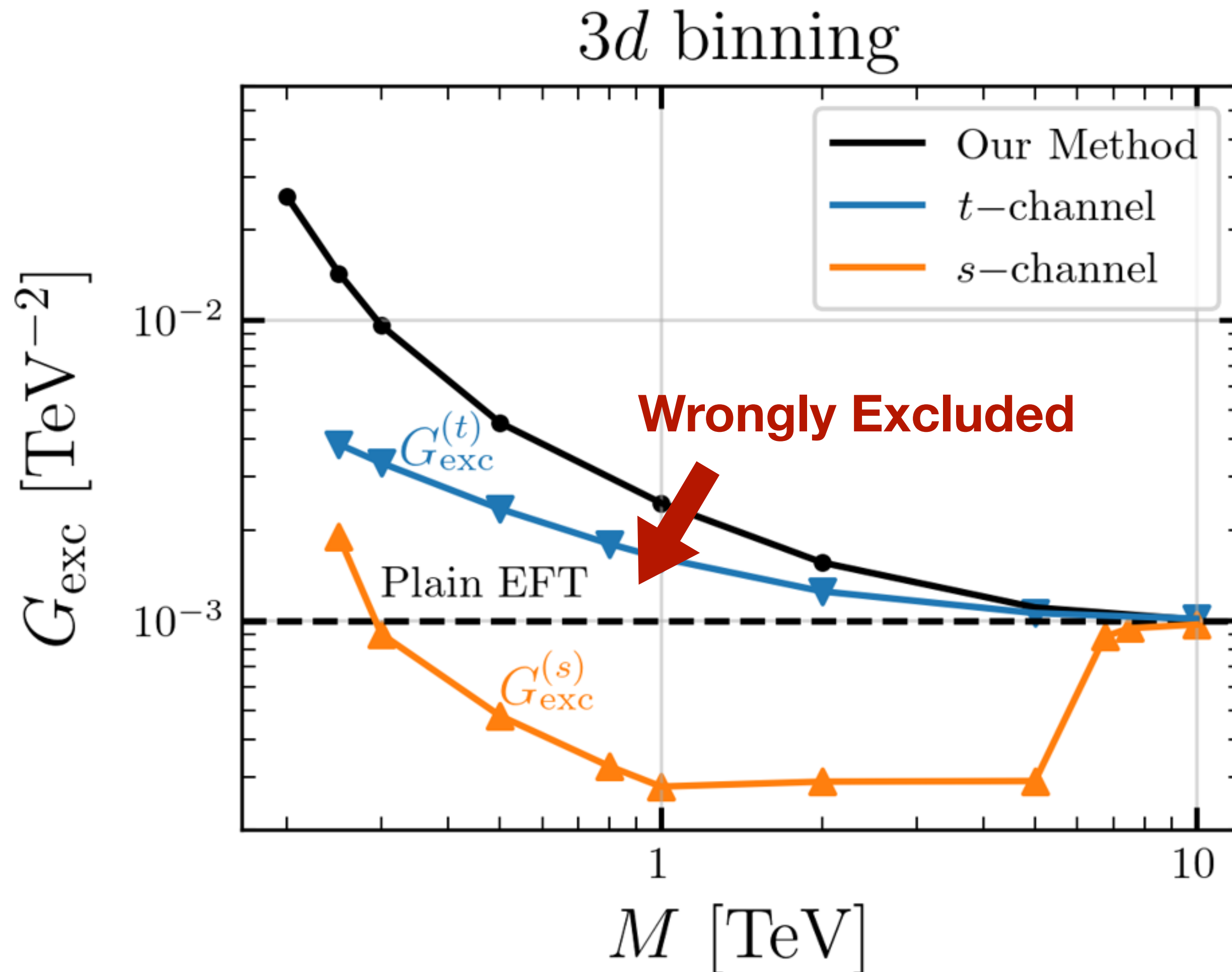
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- Plain EFT limit is stronger than t-channel UV model limit
- Our limit is weaker than UV-model limits

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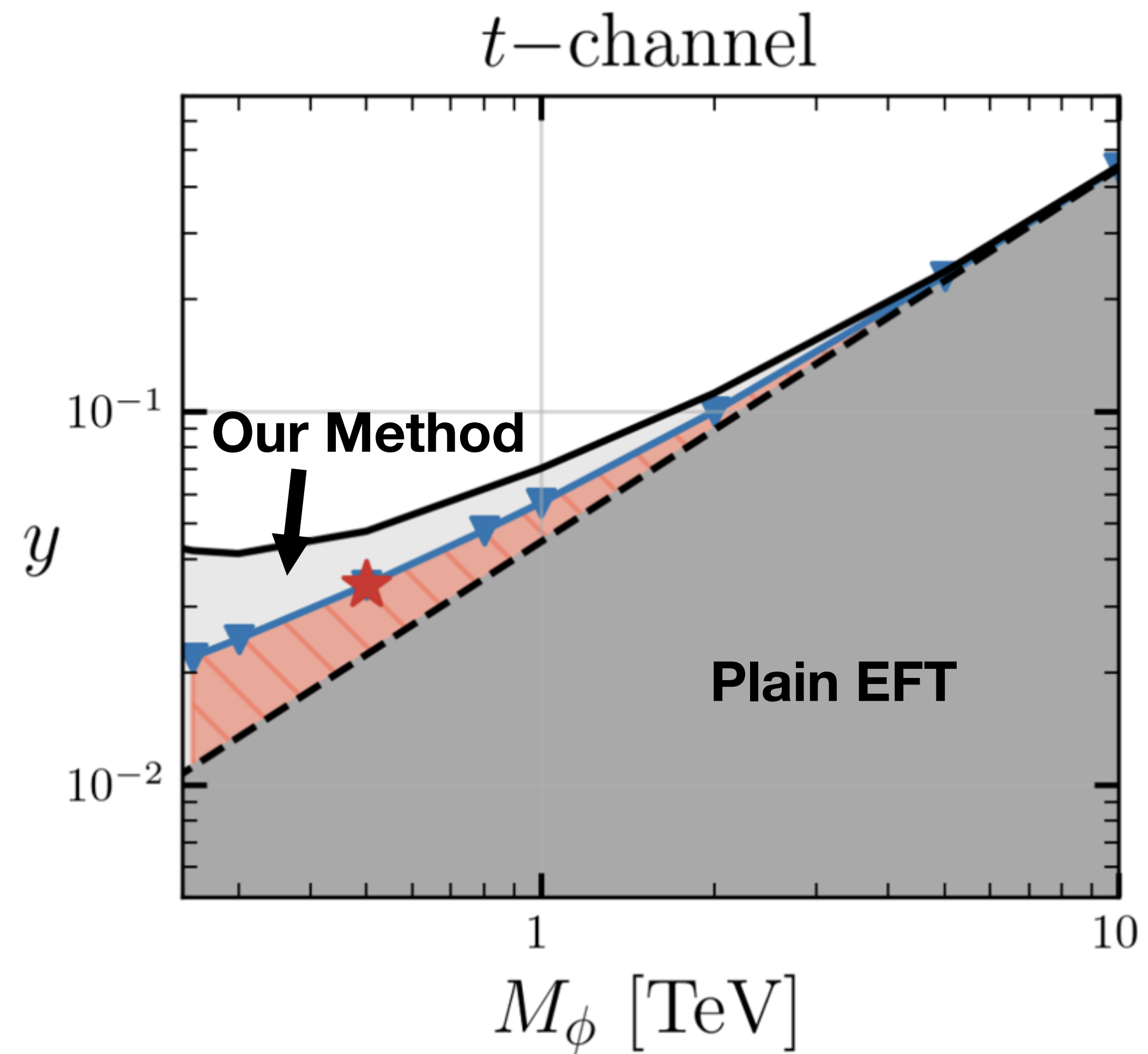
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Exclusion Limits



Open Questions: Three Point Couplings

$$\mathcal{L}_{\text{SM}} = \frac{1}{2}(\partial h)^2 - \frac{1}{2}m_h^2 h^2 - \frac{g_{hhh}}{3}h^3 + \bar{t}i\gamma^\mu\partial_\mu t - \frac{y_t}{\sqrt{2}}(v+h)\bar{t}t + \dots, \quad \mathcal{L}_{\text{BSM}} = -\frac{1}{3}G_{hhh}h^3$$

↓ **Theory Uncertainties**

$$\mathcal{L}_{\text{BSM}} = -\frac{1}{3}G_{hhh}\left[h^3 + \frac{c_1}{M^2}h^2\Box h + \frac{c_2}{M^4}h^2\Box^2 h + \frac{c_3}{M^4}h(\Box h)^2 + \dots\right]$$

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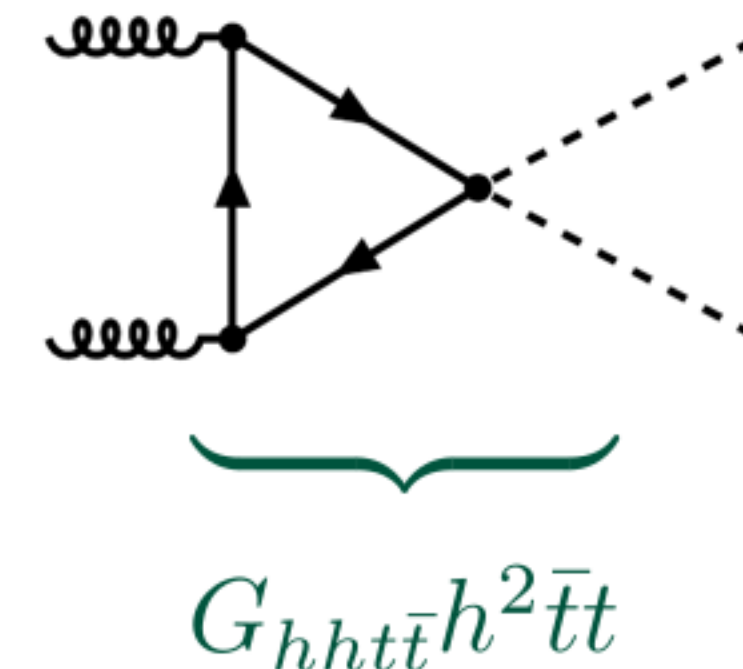
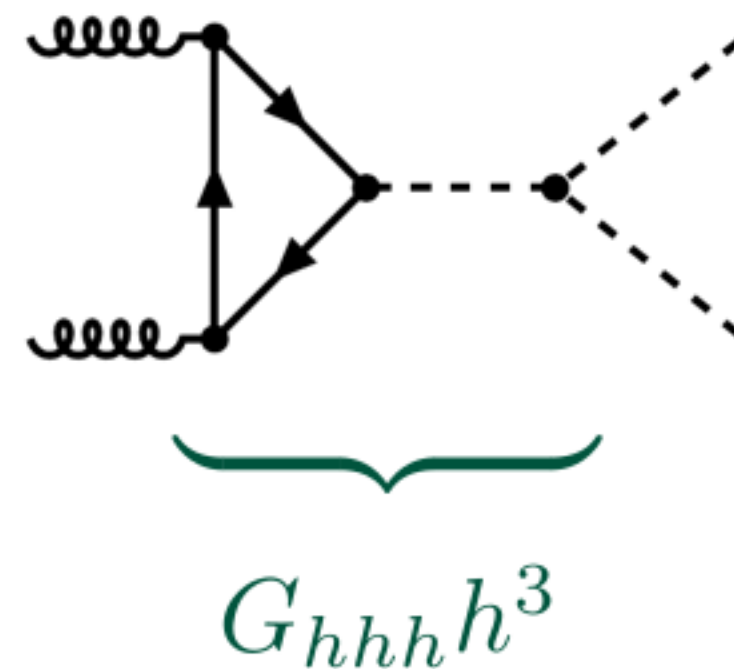
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$$\Box h \sim \bar{t}t$$

EOM



Conclusions

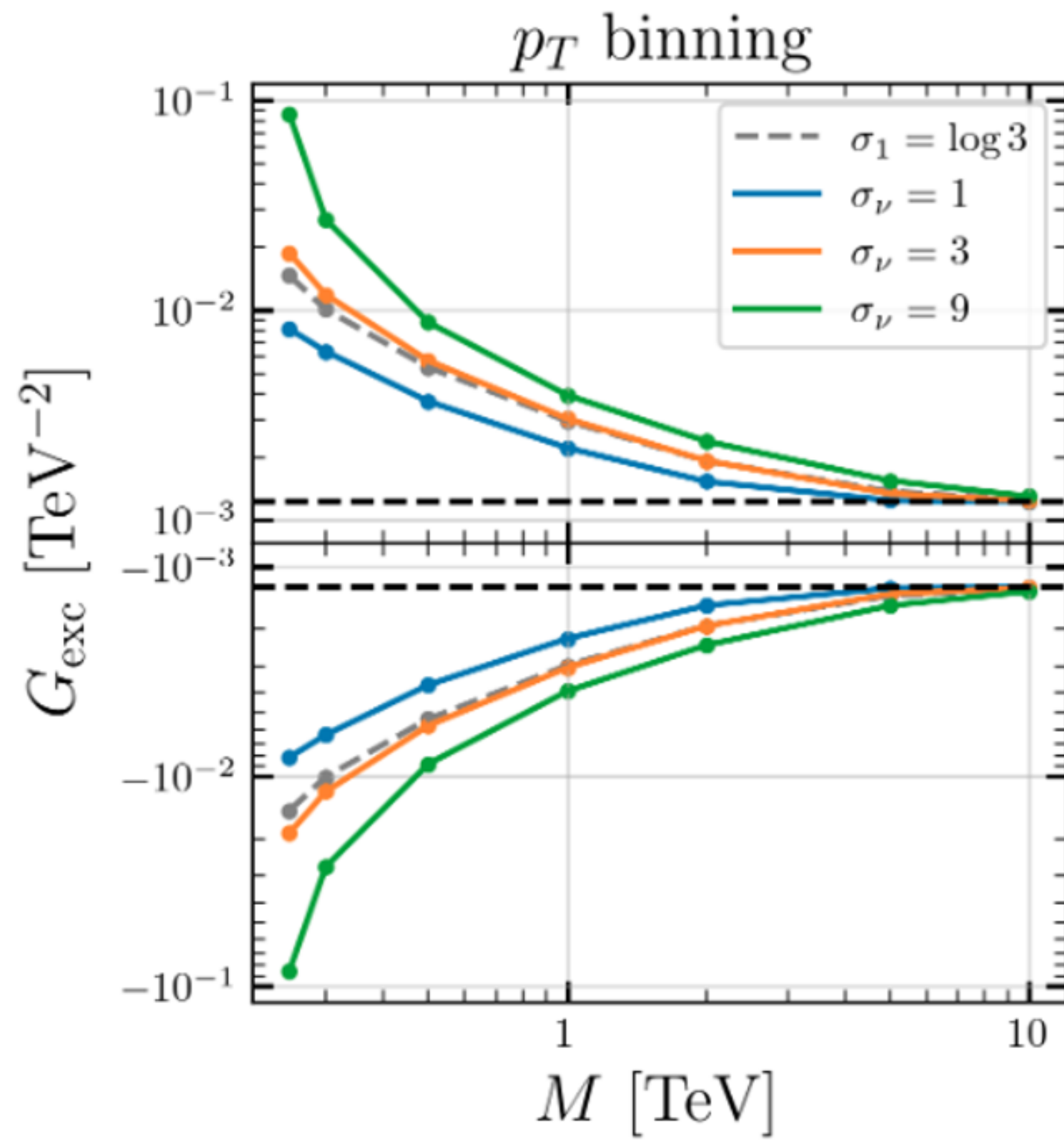
- Modeling EFT uncertainties is **the solution** to EFT validity problem

Our proposal:

- Well defined **statistical methodology** that makes explicit **prior assumptions**
- Precise notion of cutoff M enables “final” state” combinations
- Can be **easily implemented** with **event reweighting**

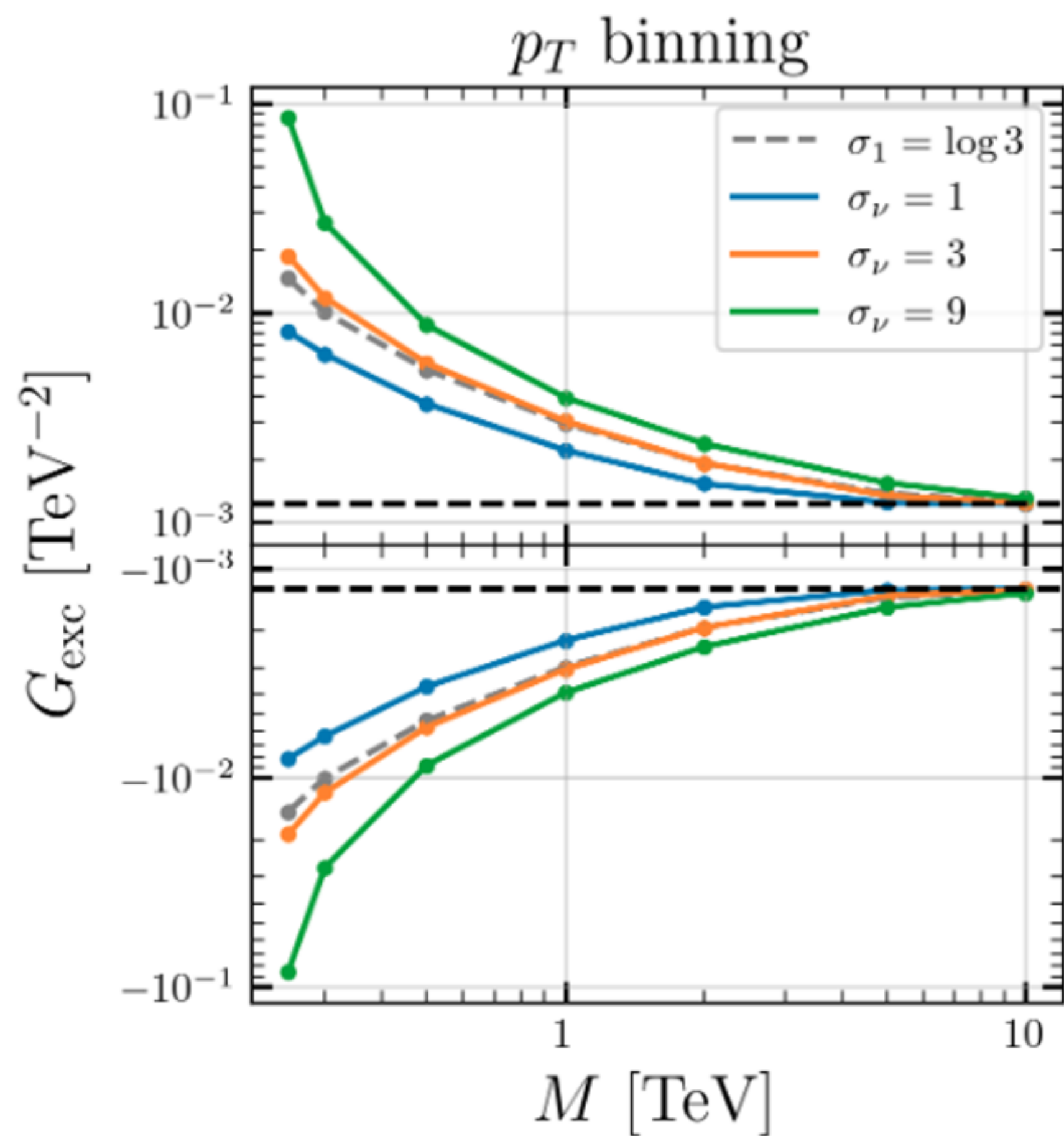
Backup Slides

Dependence on the Prior



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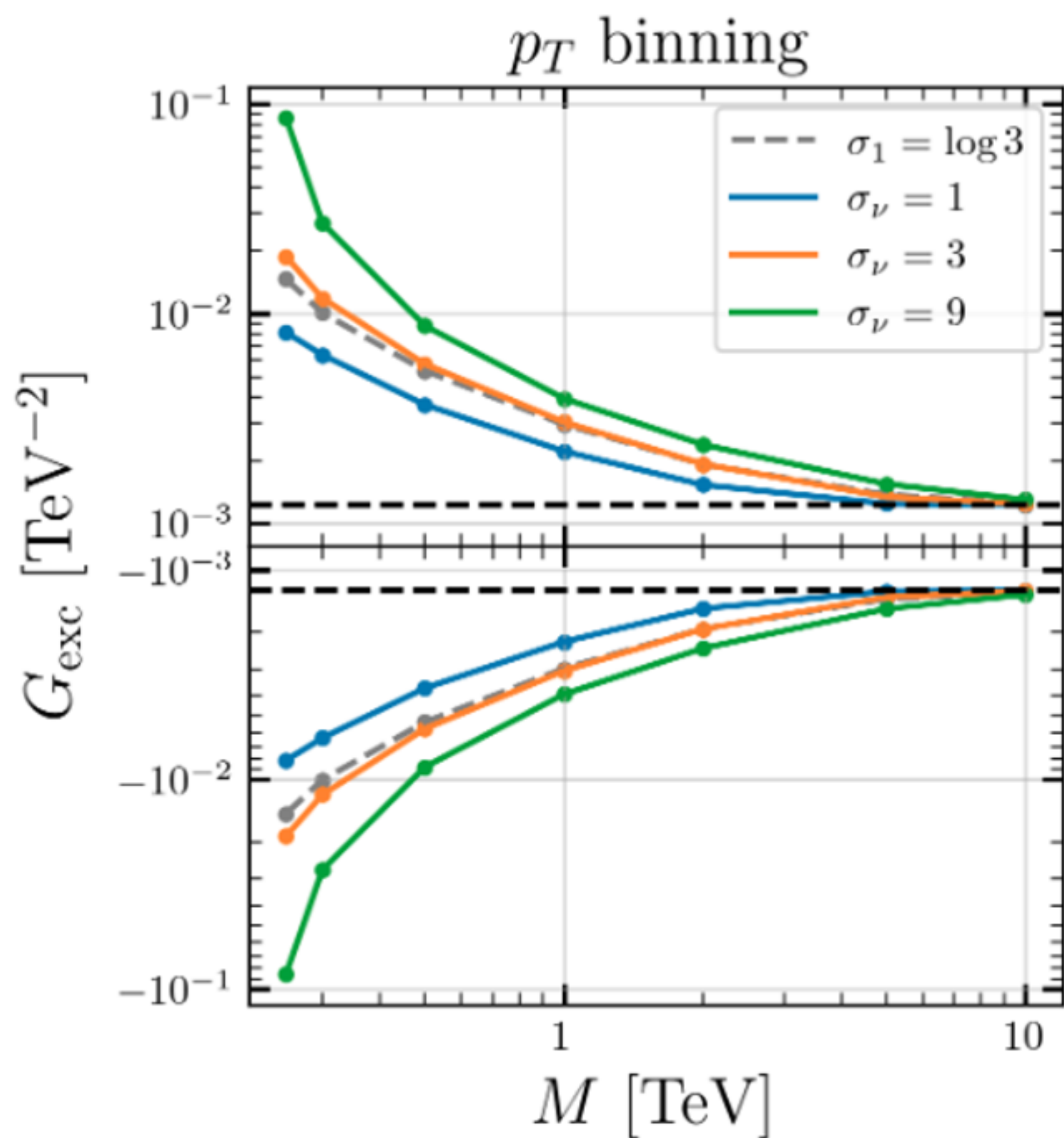
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$$\text{Prob}(|\nu| < 3) = 0.68$$

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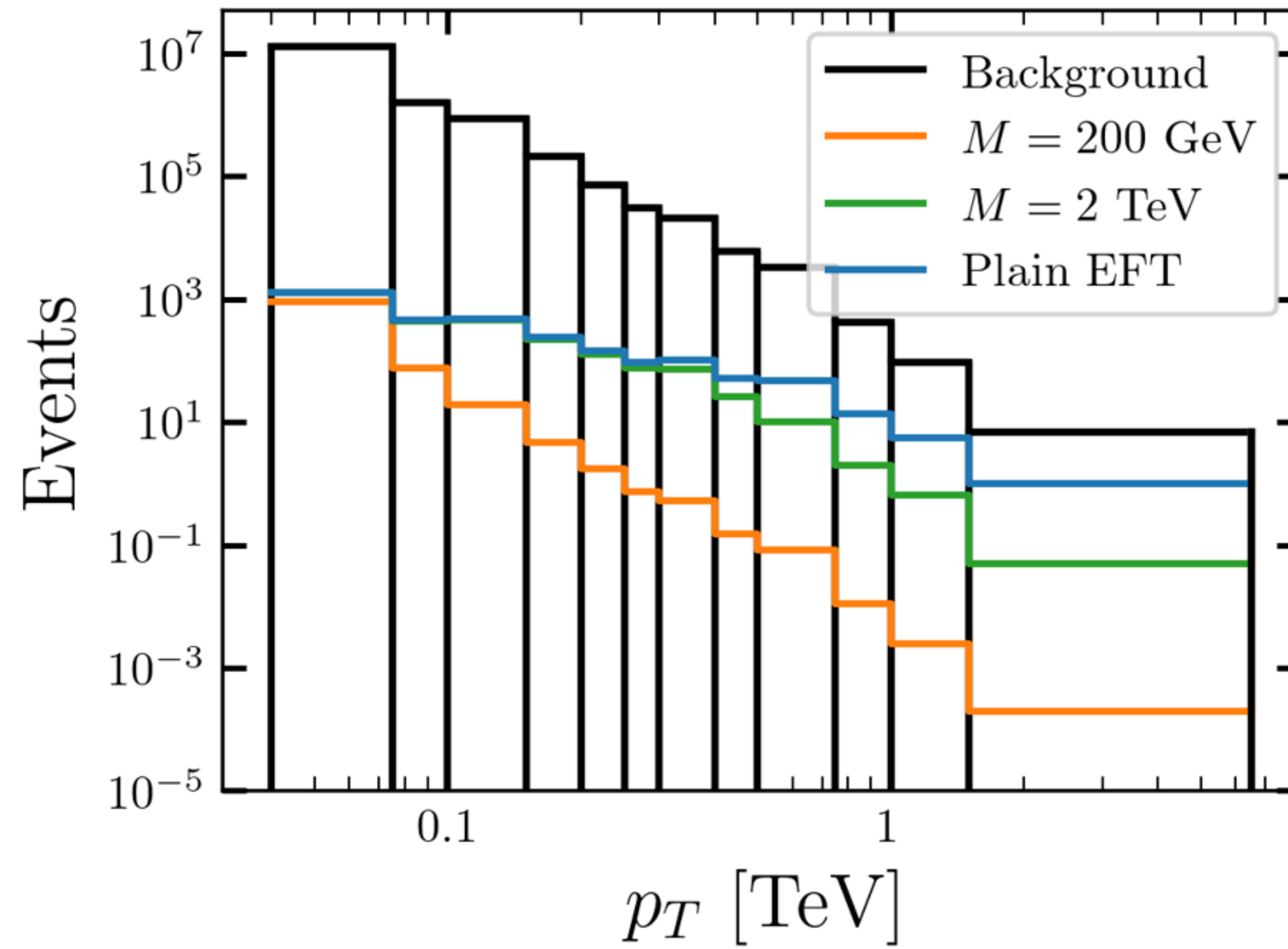


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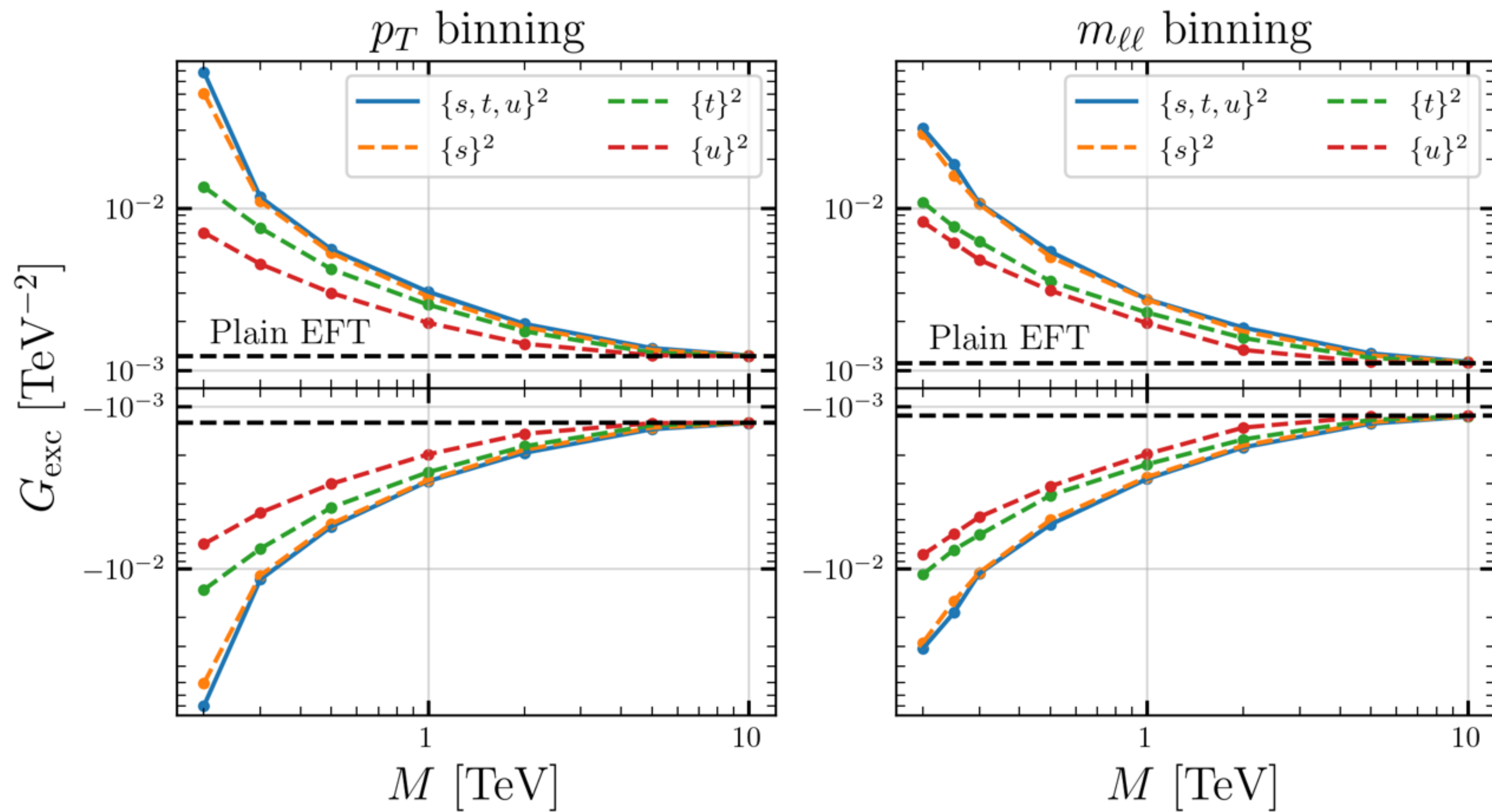
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- **Predictivity** requires bounding **higher-order** EFT terms
- **Different priors** correspond to **different physical assumptions**

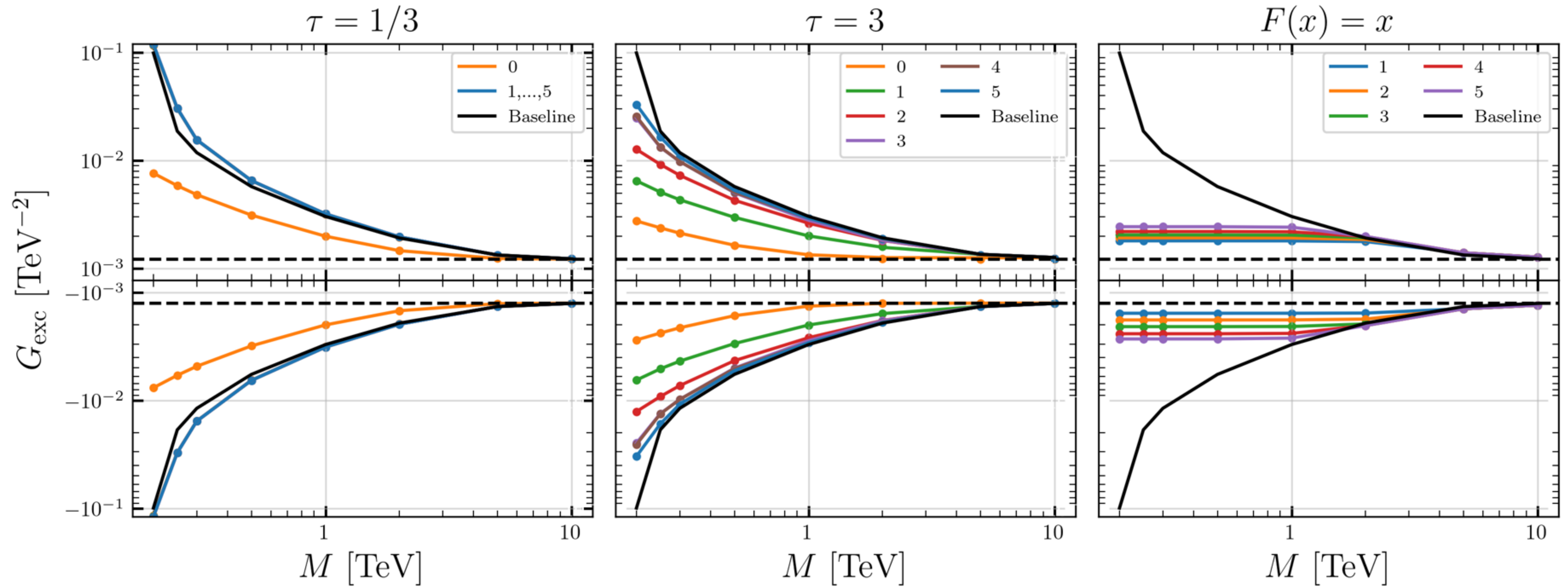
Nuisance Parameters Cancellation



s, t, u Nuisance Parameters



Different Form Factor Choices



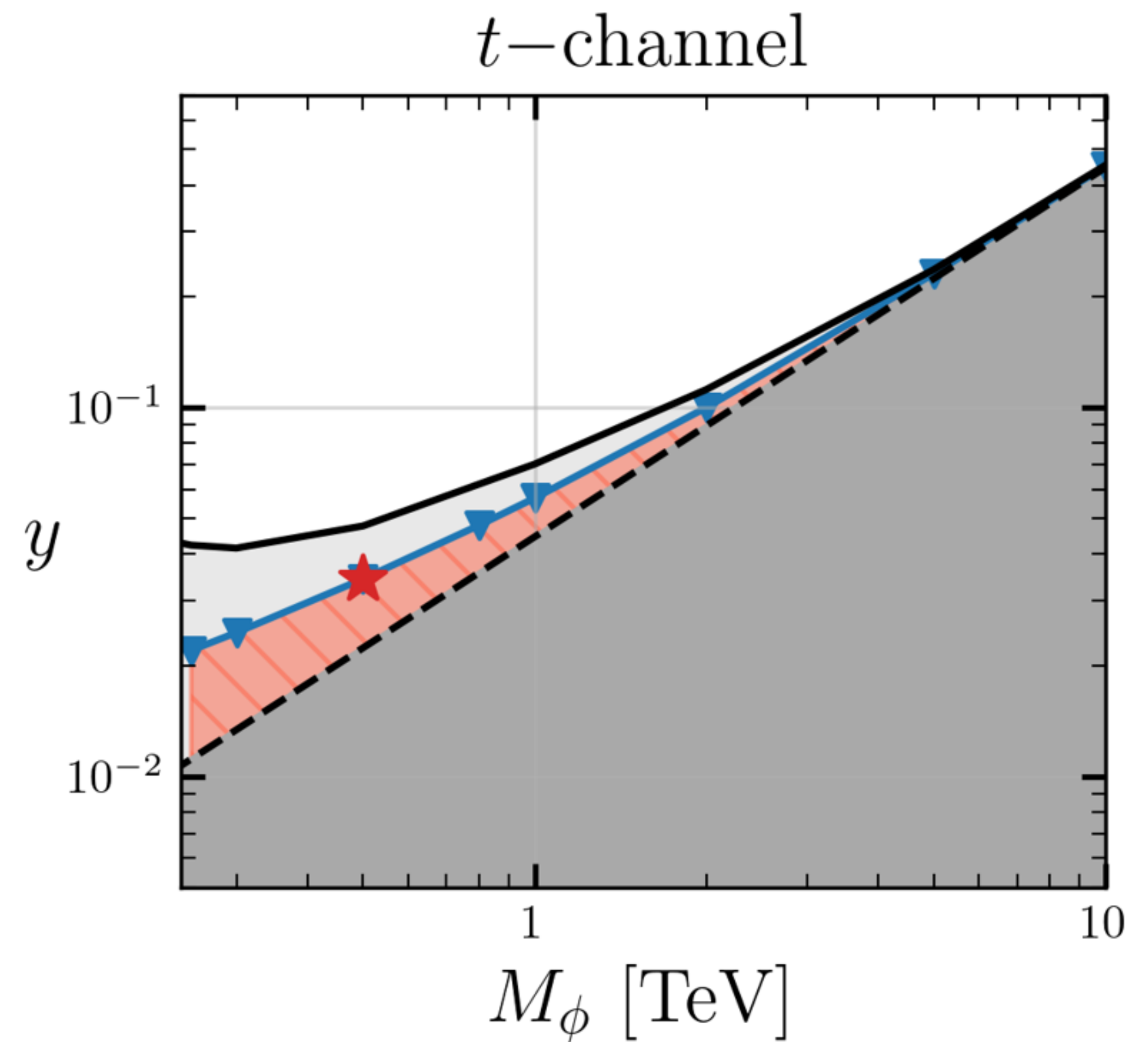
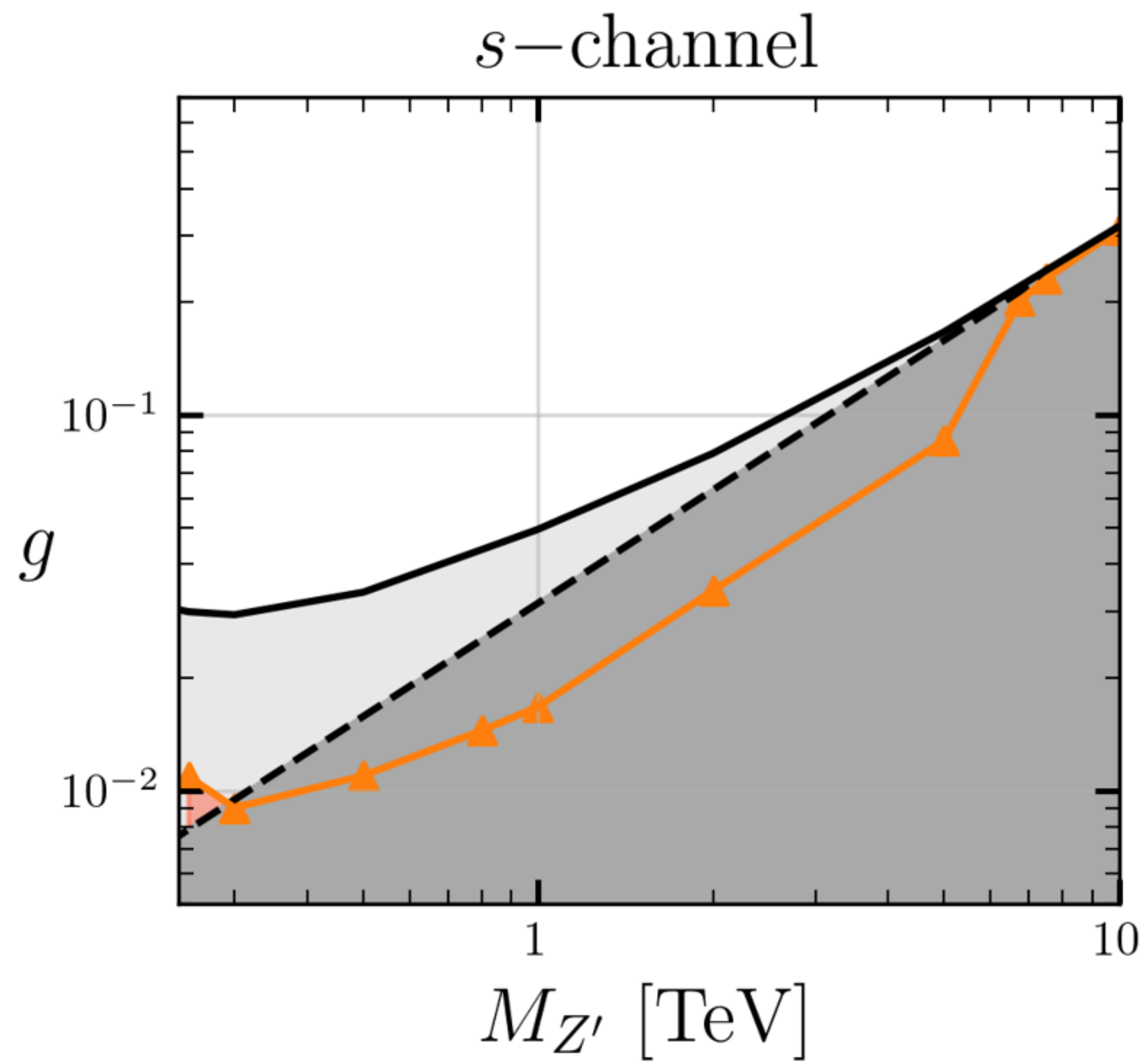
Reweighting Procedure

The Feynman amplitude depends **linearly on** G , so the weight is a quadratic polynomial

$$w_e^{\text{EFT}}(G) = w_e^{\text{SM}} [1 + \mathfrak{l}_e G + \mathfrak{q}_e G^2]$$

where w_e^{SM} are the event weights in the SM, normalized to $\sum_e w_e^{\text{SM}} = \sigma^{\text{SM}}$.
The linear and quadratic coefficients $\mathfrak{l}_e$ and $\mathfrak{q}_e$ are computed by **reweighting the MC sample** with two different nonzero values of G

Exclusion Limits



Open Questions: Three Point Couplings

$$\mathcal{L}_{\text{SM}} = \frac{1}{2}(\partial h)^2 - \frac{1}{2}m_h^2 h^2 - \frac{g_{hhh}}{3}h^3 + \bar{t}i\gamma^\mu\partial_\mu t - \frac{y_t}{\sqrt{2}}(v+h)\bar{t}t + \dots, \quad \mathcal{L}_{\text{BSM}} = -\frac{1}{3}G_{hhh}h^3$$

We could incorporate **higher order EFT corrections** to the Higgs cubic coupling similarly to what done for four-point interactions

$$\mathcal{L}_{\text{BSM}} = -\frac{1}{3}G_{hhh}\left[h^3 + \frac{c_1}{M^2}h^2\Box h + \frac{c_2}{M^4}h^2\Box^2 h + \frac{c_3}{M^4}h(\Box h)^2 + \dots\right]$$

However, **higher derivative corrections** to 3-point couplings can be eliminated by field redefinitions, effectively modifying only contact interactions.

Open Questions: Three Point Couplings

Effectively using the **equations of motion** for the Higgs field, we obtain

$$\mathcal{L}_{\text{BSM}} \rightarrow -\frac{1}{3}G_{hhh} \left[\left(1 - c_1 \frac{m_h^2}{M^2} + (c_2 + c_3) \frac{m_h^4}{M^4} \right) h^3 - \frac{y_t}{\sqrt{2}} \left(\frac{c_1}{M^2} - \frac{(c_2 + 2c_3)m_h^2}{M^4} \right) h^2 \bar{t}t - \frac{y_t c_2}{\sqrt{2} M^4} h^2 \square(\bar{t}t) \right] + O(G_{hhh}^2)$$

The h^3 form factor is equivalent to **further modification of the coupling itself**, along with the **addition of 4-point couplings**

$$\mathcal{L}_{\text{BSM}} \rightarrow -\frac{1}{3}G_{hhh}h^3 + \frac{1}{2}G_{hht\bar{t}}h^2\bar{t}t$$

Open Questions: Discovery

